the insulating metastable structure with coexisting monoclinic $C_{2/c}$ and R′ phase and show a huge ~2.2×10^4 times (highest ever to the best of our knowledge) DC switching under electric filed (see the note S3) and as a consequence the crystal structure is transformed to the R phase.

Temperature dependent V-I characteristics and threshold field for IMT

Fig. 3a presents the isothermal V-I characteristics at some selected temperatures. Detailed hysteresis behaviors are shown in Fig. S6 and discussed in note S5. A giant DC (2.2×10^4 %) switching is observed at 7K and its value gradually reduces with increasing temperature (shown in Fig. 3b). Interestingly, the HRS to LRS occurs only after a threshold voltage value V_{th} which decreases with increasing temperature (shown in Fig. 3c). The required threshold field (V_{th}/d) (15 V/cm at 300K) in our V_2O_3/Si thin film is at least two order smaller than the other prototypical materials like AM_4Q_8 (A- Ga, Ge; M - V, Nb, Ta; X - S, Se), $NiS_{2-x}Se_x$ and $V_{2-x}Cr_xO_3$ [35,36,37].

Temperature dependent Raman measurement under zero and finite electric field

To understand the thermal evolution of landscape of structural phase coexistence and effect of electric field on it, we performed T dependent Raman studies with and without application of external electric field. Fig. 4a represents the T-dependent Raman spectra without external electric field, which shows the onset of a structural phase transition from rhombohedral R′ (R-3c) to $C_{2/c}$ monoclinic phase at 150K (T_{SPT}) though the system remains in the insulator state and this phase transformation is not complete even up to 7K (shown in Fig. S5a). Both the phases coexist below 150 K, though monoclinic phase dominates over rhombohedral phase as the temperature is lowered. Interestingly, under the application of 100 mA current, the SPT is suppressed (shown in Fig. 4b) and the Raman spectra corresponding to the room temperature R phase persists even up to 7K. This is similar to our R vs T results under 100mA, where the room temperature electronic state is arrested and it remains in the LRS throughout the measured temperature window (Fig 1i).

Room temperature Mott field effect transistor

As shown in Fig. 1i a giant electric field induced conductance change at room temperature is observed in the present V_2O_3/Si film. This property can be utilized to fabricate a Mott FET (MFET). Apart from direct applications, such Mott transistor is used for confirmatory test about the mechanism involved in electric field induced IMT. If the IMT could indeed be triggered solely by electron injection on application of an electric field, then only resistive switching is possible in a three-terminal FET geometry with a high-k gate dielectric eliminating the leakage current. To check this, we fabricated a Mott FET transistor where the channel resistance in V_2O_3 is controlled by gate voltage. Schematic of the three terminal device structure is shown in Fig. 5a. The V-I characteristics of V_2O_3 channel are measured under different gate voltages (Vg) like (0→0.9V→0V→1.5V→0V) and plotted in Fig. 5b. For Vg = 0 V, the current is negligibly small, which can be considered as OFF state. However, on application of gate voltages (0.9 V, 1.5 V), the conductivity of the V_2O_3 channel increases significantly resulting in high current flow. This state can be considered as ON state. The channel conductance is plotted in Fig. S8. Because of very high dielectric gate insulation the leakage current is very small (~100 nA). On application of gate voltage, the insulator channel is converting to metallic state through Mott mechanism driven by electron injection making the channel a LRS. This result confirms that the HRS to LRS transition is driven by purely electronic Mott transition and

rules out the possibilities of Joule heating. Notably this MFET requires very low power in comparison to the existing transistors.

Discussions

It is well known that the systems stabilized near the phase boundary can be switched to the other phase by small external stimuli and provides higher response. For example, high electromechanical response in relaxor ferroelectric is observed in systems stabilized by doping induced disorder near to the morphotropic phase boundary [38,39]. In present system, by controlling the disorder and defects, the V_2O_3/Si thin film is stabilized at metal–insulator phase boundary (for *c*/*a* ~2.79) [14] which resulted in large resistive switching at room temperature.

In V^{3+} valance state, two unpaired electrons belong to the 3d t_{2g} orbital and because of the trigonal distortion, t_{2g} orbital further splits into a_{1g} and e_g^{π} orbitals, and the spiting energy can be tuned by changing the crystallographic *c/a* ratio [40]. Upon pressure [15], Cr doping [41,42,43] and epitaxial strain [14], the *c/a* ratio can be decreased leading to a shorter V-V separation which causes a higher a_{1g} (d_{Z^2}) orbitals overlap resulting in an increase in the coulomb repulsion energy. Because of this enhancement in coulomb repulsion, insulating state can be stabilized with reduced *c/a* ratio.

Two-orbital Mott insulator model [29] predicted that within a metal-insulator coexisting region, the gap separating two bands of different orbital character can be collapsed by reducing the orbital polarization "m" (= N_1-N_2, i.e. population imbalance between the lower and upper orbitals). This orbital polarization can be tuned by application of electric field. G. Lantz et al [44] showed that the increase in the electron population in a_{1g} electronic band (which reduces the orbital polarization) reduces the separation between a_{1g} and e_g^{π} bands. Therefore, when we apply electric field, the electron population in a_{1g} band is increased through electron injection which decreases the a_{1g}- e_g^{π} splitting and collapses the Mott gap above a threshold voltage V_{th}, which drives the HRS to LRS phase transition because of electron-electron correlation effect [29,45,46]. A schematic band diagram is shown in Fig. 5c,d. As a consequence of smaller separation between a_{1g} and e_g^{π} state, V-V separation increases which favors larger *c/a* ratio [47] that leads to an isostructural R′ to R phase transition under application of external electric field (shown in Fig. 5e). Raman mode A_{1g} represents the V atoms vibration along the *c*-axis and thus is highly sensitive to the V-V separation along *c*-axis [,44,47,48]. Therefore, A_{1g} mode can be used to probe the trigonal distortion [49]. Softening of the A_{1g} mode with external electric field signifies the increase in V-V distance and confirms the crystal structure transformation to the R phase. That is also responsible for the suppression of structural phase transition as well as huge resistance change with lowering of temperature. Once LRS phase stabilizes because of the electron injection in the a_{1g} band under external field at room temperature, the gap collapsed state is maintained as long as the electron is populated in the a_{1g} band. Therefore, the room temperature LRS- R phase is arrested under field even when the sample temperature is lowered to 7K. This is a possible mechanism of arresting the high temperature R phase under field cool cycle. Here, the electron injection is the driving force and clearly establishes that the electron-electron correlation is the main factor for the IMT in the V_2O_3.

The effect of electric field is to increase the electronic concentration in a_{1g} band beyond the critical value (V_{th}) destabilizing the insulating state and triggering to a IMT. Since field induced carrier concentration increases with temperature, the switching voltage V_{th} is reduced with temperature as shown in Fig. 3c. Notably the required V_{th} for IMT is remarkably small, <1V. The effect of disorder/defects is like Cr doping, it reduces the *c/a* ratio [31,49] and therefore, the potential barrier between two competing phases is reduced. As a matter of fact, in sample with higher defect densities, more carriers can be generated with less electric

field or power [11,29,36,[50]]. For an optimal *c/a* ratio, the system can be stabilized at the HRS-LRS phase boundary and a very small external perturbation like electric field can trigger the IMT. Therefore, room temperature low power resistive switching can be achieved by controlling defect densities as observed in our V_2O_3/Si thin film and its multistate resistive switching in two terminal device configuration would be a potential candidate for the neuromorphic computing.

Realization of resistive switching in FET geometry at room temperature confirmed that the IMT is electron-electron correlation driven. The very small leakage current rules out the joule heating effect in the IMT. Here we showed 15 times ON/OFF resistivity ratio using Vg =1.5 volt which would be a step towards the practical application of MFET as a low power device. However, more device optimization is needed to reach its full potential.

Conclusion

For applications, resistive switching mechanism compatible with Si is highly sought and in this study, resistive switching at room temperature is demonstrated in V_2O_3 polycrystalline films deposited on Si. It is shown that by controlling defects a phase boundary compound (film) with desired c/a ratio can be stabilized. Further, a three terminal test device mimicking Mott FET is fabricated and resistive switching is controlled through gate voltage. This device showed 15 times conductivity change in film at room temperature by application of small gate voltage which opens the possibility to fabricate correlated oxide based low power Mott transistor. The study unequivocally showed that the electron injection mechanism is responsible for the Mott insulator to metal transition along with isostructural transition in V_2O_3/Si film. Electric field induced conductivity along with crystal structure change in linear four probe geometry and FET geometry confirm that the Mott gap collapse is the key mechanism of the IMT.